1. Clinical Question & Core Hypothesis

Clinical Question

High tumor mutational burden (TMB) should increase neoantigen load and sensitize tumors to immune checkpoint therapy, yet many NMIBC tumors with elevated TMB fail to respond. Why does high TMB not reliably translate into effective anti-tumor immunity in NMIBC? 2) Tier 1 paths used: I built on the TMB path (links neoantigen load to immunogenicity) and the Immunoscore/Immunotherapy-biomarker paths (spatial immune contexture as the effector readout that mediates response). These provide a mechanistic backbone: tumor-intrinsic antigenicity (TMB) must be converted into effective T cell responses by organized local immune architecture (Immunoscore/TLS). 3) Tier 2 evidence supporting the claim: MSH6/MMR status (affecting TMB generation) and systemic/metabolic modifiers (HDL-C, ALP, TG, SHBG, IGF1) plus epigenetic shed markers (SDC2/TFPI2 methylation) can modulate antigenicity, immune recruitment, or immune function and therefore explain heterogeneity in the TMB→response axis. 4) Why I chose therapy_response as the endpoint: The Ontologist prioritized immunotherapy biomarker and immunoscore signals; the main clinical puzzle is predicting which NMIBC patients will actually benefit from immune-targeted therapies, so a therapy_response proxy directly tests the mechanistic question and has immediate translational impact. 5) What I deliberately EXCLUDED and why: I excluded Tier3 exploratory cross-disease signals (CSF AD biomarkers, SARIFA/BRAF observations and AI-accessibility implementation ideas) from the core hypothesis because they are weakly linked mechanistically to NMIBC response biology; they remain useful for future methods work but would dilute the focused mechanistic claim?

Working Hypothesis

In NMIBC, high tumor mutational burden increases neoantigen load but effective anti-tumor immunity requires the formation of organized TLS-like lymphoid aggregates that support local antigen presentation, B/T cell collaboration, and germinal center–like reactions, thereby generating an Immunoscore-like spatial immune context that converts antigenicity into productive T cell–mediated attack. Clinically, patients with high TMB and TLS-rich NMIBC will have higher response rates and more durable control with checkpoint blockade compared with high-TMB TLS-poor or low-TMB tumors, and integrating TLS status with TMB and Immunoscore will improve predictive accuracy for immunotherapy benefit and guide patient selection in NMIBC.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	TMB, Immunoscore
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

TMB: TMB not reliably translate into effective anti-tumor immunity in NMIBC? 2) Tier 1 paths used: I built on the TMB path (links neoantigen load to immunogenicity) and the Immunoscore/Immunotherapy-biomarker paths (spatial immune contexture as the effector readout that mediates response).

Immunoscore: immunoscore signals; the main clinical puzzle is predicting which NMIBC patients will actually benefit from immune-targeted therapies, so a therapy_response proxy directly tests the mechanistic question and has immediate translational impact.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	MSH6	Supporting modifier from knowledge graph
Secondary	HDL-C	Supporting modifier from knowledge graph
Secondary	SDC2	Supporting modifier from knowledge graph
Secondary	TFPI2	Supporting modifier from knowledge graph
Secondary	ALP	Supporting modifier from knowledge graph
Secondary	TG	Supporting modifier from knowledge graph
Secondary	SHBG	Supporting modifier from knowledge graph
Secondary	IGF1	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

Prediction 2: At the tissue level, tumors with the imaging feature but low anchor gene activity will show immune cells confined to structural peripheries rather than infiltrating tumor nests.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- MSH6: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- HDL-C: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.